Self-purging inflation valve core with a deflectable seat ring — Prosecution Assessment

Likelihood of grant	6 / 10	Claim 1 is not disclosed by any saved reference, but the only built embodiment is described with a mechanism that cannot work as written and its layout must come from the inventor before filing; scored on the application as reviewed in round 1, so fixes made in the later fix pass are not re-scored, and the score rests only on the 17 references we found and saved.

Deficiencies and prior art the application will face

Status key: `open, routed to fix pass` = upheld and sent to the fix pass running in parallel (not re-reviewed); `open` = upheld or unclear and not routed; `prior art` = a saved reference. Rows marked *(unclear)* turn on facts only the inventor or counsel can supply.

#: 1 · **Deficiency or prior-art reference:** The description never explains how the purge shoulder reaches the seat ring. The poppet opens towards the chamber ([0067]), so a shoulder "on the chamber side of the sealing face" ([0059], claim 3) moves away from the ring as it opens, and no ring surface for it to press on is described. · **Type:** Enablement · **Claims affected:** 3, 4, 22 · **Severity:** high · **Source:** F-32-20 · **How to respond:** Get the axial order of the parts (internal shoulder, sealing edge, convolutions, purge shoulder) and the ring face contacted from the inventor's sketch or prototype; add one paragraph; correct [0138] and claim 3 if reversed. This cannot be added after filing. · **Status:** addressed in fix pass (not re-reviewed)

#: 2 · **Deficiency or prior-art reference:** The same layout gap seen from the ring bore: the 2.3 mm bore admits only the poppet stem ([0110], [0111]) and [0138] puts the sealing edge at the supply-side end, so the preferred form of claims 3 and 4 cannot be built from the words, and there are no drawings to help. · **Type:** Enablement · **Claims affected:** 3, 4, 8 · **Severity:** high · **Source:** F-gate14-2 · **How to respond:** As row 1: one layout paragraph from the inventor's FIG. 1 sketch, drawn; conform [0042], [0059], [0134] and claim 3 if the shoulder sits elsewhere. · **Status:** addressed in fix pass (not re-reviewed)

#: 3 · **Deficiency or prior-art reference:** For the illustrated core, no text says which ring surface takes the poppet's load, or how purge air in the outer clearance gets past the press-fit band at the internal shoulder to the chamber side ([0057], [0075], [0078], [0081]). The figure plan itself calls that arrangement a drafting assumption. · **Type:** Enablement · **Claims affected:** 1, 4, 21, 30, 31, 58, 79 · **Severity:** high · **Source:** F-28-3 · **How to respond:** Inventor to state the loaded surface and the purge exit route; add them to [0059], [0074], [0078], [0081]; reconcile [0101], [0127] and [0138]. No claim narrowing. · **Status:** addressed in fix pass (not re-reviewed)

#: 4 · **Deficiency or prior-art reference:** Same gap, as it bears on claim 1's through-flow for the FIG. 1 core. Claim 1 itself survives on the coherent inner-surface arrangement of [0127], but claims 4, 21 and 30 read directly on the unexplained embodiment. · **Type:** Enablement · **Claims affected:** 1, 4, 21, 30, 58, 79 · **Severity:** high · **Source:** F-36-1 · **How to respond:** As row 3: state the axial arrangement and the purge route (past the free end, through purge ports 262 or flutes 362) from the inventor's sketches. · **Status:** addressed in fix pass (not re-reviewed)

#: 5 · **Deficiency or prior-art reference:** The built core is said to open its purge gap by deflecting outward ([0041], [0075]), yet the gap lies between the ring's outer surface and the bore ([0061]). Moving outward closes that clearance, as [0125] itself says. The wiper ring has the same contradiction ([0115], [0185]). · **Type:** Enablement · **Claims affected:** 9, 29, 30, 34, 72 · **Severity:** high · **Source:** F-28-2 · **How to respond:** Inventor to state the real movement of prototypes P-1 to P-3; rewrite [0041], [0061], [0075], [0076], [0115], [0185], the flutes

of [0132] and the FIG. 1, 3, 6 and 10B instructions to it. No claim narrowing. · **Status:** addressed in fix pass (not re-reviewed)

#: 6 · **Deficiency or prior-art reference:** Same contradiction, as it bears on Example 1 and on claims 29 with 30 and claim 72. The only tested embodiment is described with a mechanism that cannot work as written, which weakens its value as evidence. · **Type:** Enablement · **Claims affected:** 1, 9, 29, 30, 72 · **Severity:** high · **Source:** F-30-6 · **How to respond:** As row 5. If the movement is inward or ovalising, amend "outward" in the illustrated embodiment; claim 1 is not narrowed. · **Status:** addressed in fix pass (not re-reviewed)

#: 7 · **Deficiency or prior-art reference:** Same contradiction, plus the "free" part of the ring's outer surface ([0057], [0075], [0125]) reads as a clearance at rest, against [0061] "the purge gap 60 is closed". · **Type:** Enablement · **Claims affected:** 9, 29, 30, 72 · **Severity:** high · **Source:** F-gate14-4 · **How to respond:** As row 5; use the wording of [0125] if the gap opens by inward movement; state one at-rest condition. · **Status:** addressed in fix pass (not re-reviewed)

#: 8 · **Deficiency or prior-art reference:** *(unclear)* The test does not show that air pressure, rather than the pump head pressing on the poppet, caused the over-travel. The snap was heard only when the pump was removed ([0181]), and travel and pressure were not measured ([0179], [0188]). If the head pressed the poppet, the tested product falls outside claims 1, 58 and 79. · **Type:** Enablement · **Claims affected:** 1, 58, 79, 83, 84, 87 · **Severity:** high · **Source:** F-32-7 · **How to respond:** Ask the inventor for the pump-head type and whether it touches the poppet tip, whether a snap occurred between strokes, and any pressure trace; correct [0065], [0090], [0181], [0202] to the answer. · **Status:** open

#: 9 · **Deficiency or prior-art reference:** The dimension and pressure ranges are stated to belong to the superelastic ring with a convoluted wall ([0110], [0113], [0116], [0168], [0184]) but are claimed for any material and wall, including a PEEK ring (claims 76–77) and a ring of no stated material (claim 67). · **Type:** Support · **Claims affected:** 13, 16, 17, 35, 41–44, 57, 67, 69, 70, 76, 77, 82 · **Severity:** high · **Source:** F-28-1 · **How to respond:** Ask the inventor whether the ranges hold for PEEK, resilient metal and other walls. If yes, say so and delete the self-limiting sentences; if not, add "superelastic alloy" to claim 67 and remove the ranges from claims 76–77 (narrowing: counsel decides). · **Status:** addressed in fix pass (not re-reviewed)

#: 10 · **Deficiency or prior-art reference:** No drawings exist. The package holds only a figure plan, which also contains internal working notes that must never be filed. Filing without drawings risks re-dating or loss of the figure references at the EPO (Rule 56) and a drawings objection in the US (35 U.S.C. 113). · **Type:** Formal · **Claims affected:** all · **Severity:** high · **Source:** F-36-2 · **How to respond:** Commission FIGS. 1–15 once rows 1–7 are answered; add a Brief Description of the Drawings; strip all working notes from the instructions. · **Status:** addressed in fix pass (not re-reviewed)

#: 11 · **Deficiency or prior-art reference:** No abstract, no brief description of the drawings, and no technical field, background or summary; [0183] points to a background that does not exist. · **Type:** Formal · **Claims affected:** — · **Severity:** high · **Source:** F-gate14-1 · **How to respond:** Supply the missing sections (abstract and list of figures at assembly); re-point [0183] to the definitions at [0045] and [0050]. · **Status:** addressed in fix pass (not re-reviewed)

#: 12 · **Deficiency or prior-art reference:** Lack of unity: claims 67 (seat ring) and 90 (making it) share with claim 1 only a ring that deflects radially and recovers, which US 9982818 B2 already shows in wave form. There are also three product independents in one category (1, 58, 67), which needs a Rule 43(2) justification at the EPO. · **Type:** Formal · **Claims affected:** 1, 58, 67, 79, 90 · **Severity:** high · **Source:** F-30-7 · **How to respond:** Tie claims 67 and 90 to the purge function (superelastic alloy, or the matched kit of [0163]), or divide them out. Expect a US restriction requirement. · **Status:** addressed in fix pass (not re-reviewed)

#: 13 · **Deficiency or prior-art reference:** Claim 67 (the seat ring alone) is a known wave-section ring (US 9982818 B2) made of superelastic NiTi (US 6393765 B1) and sized for a Presta bore. It recites no poppet, over-travel or purge gap. It is not anticipated, but the obviousness combination is credible. · **Type:** Inventive step · **Claims affected:** 67–72, 76–78 · **Severity:** high · **Source:** F-32-1 · **How to respond:** Narrow: add superelastic alloy with Af below 5 °C and a continuous sealing edge, or recast as the matched kit of [0163]. · **Status:** addressed in fix pass (not re-reviewed)

#: 14 · **Deficiency or prior-art reference:** Claim 90 (making the ring) is the standard NiTi tube route: laser cut, shape-set in a fixture, quench, electropolish ([0169]–[0173]). It states no process effect, so it stands or falls with the ring. No saved reference documents this route; an examiner would cite general NiTi processing art. · **Type:** Inventive step · **Claims affected:** 90, 92, 94–98 · **Severity:** high · **Source:** F-32-2 · **How to respond:** Tie claim 90 to the ring features that survive row 13, or to making the claim 1 valve core with the press-in step of claim 100. · **Status:** addressed in fix pass (not re-reviewed)

#: 15 · **Deficiency or prior-art reference:** Every independent claim requires the purge to happen because pressure pushes the poppet into over-travel. The description itself discloses three other routes a competitor can use freely: over-travel by a depressor or by hand ([0088]), a separate purge actuator ([0145]), and Schrader cores pressed by the pump head ([0159]). In the US, matter disclosed but not claimed risks being dedicated to the public. · **Type:** Design-around · **Claims affected:** 1, 46, 57, 58, 79 · **Severity:** high · **Source:** F-32-4 · **How to respond:** Define the purge state by poppet position beyond the normal stroke, with the pressure trigger and mechanical depression as dependent alternatives; or add an independent claim with a pressure-responsive purge actuator. · **Status:** addressed in fix pass (not re-reviewed)

#: 16 · **Deficiency or prior-art reference:** Same escape for method claim 79, which requires "raising the pressure differential", and for deflation-only purging ([0089]). Against claims 1 and 58 a defendant uses [0080] and the word "additionally" in claim 46 to argue that a pressure-driven purge is required. · **Type:** Design-around · **Claims affected:** 1, 46, 58, 79, 88 · **Severity:** high · **Source:** F-33-4 · **How to respond:** As row 15: a position-defined purge state, with pressure, depressor and deflation as dependents ([0033], [0088], [0089]). · **Status:** addressed in fix pass (not re-reviewed)

#: 17 · **Deficiency or prior-art reference:** The "normal-open" state requires the ring to be "in its at-rest form", so a design in which the poppet lightly touches the ring before the threshold escapes. Claim 40 (threshold set by ring stiffness, [0144]) also conflicts with claim 1, because there the ring is loaded below the threshold. · **Type:** Design-around · **Claims affected:** 1, 40, 58, 79 · **Severity:** high · **Source:** F-33-2 · **How to respond:** Delete "the deflectable ring element is in its at-rest form and" from the normal-open state (broadening, supported by [0064], [0124]); amend [0140] and [0144] only if the inventor confirms that sub-threshold contact was conceived. · **Status:** addressed in fix pass (not re-reviewed)

#: 18 · **Deficiency or prior-art reference:** The wipe-only version ([0087]: ring sealed at its edge, cleaning by the snap-back wipe alone) is outside every independent claim. The test found that the wiped surfaces were the clean ones ([0190]). · **Type:** Design-around · **Claims affected:** 1, 58, 67, 79 · **Severity:** high · **Source:** F-gate14-5 · **How to respond:** Add an independent core claim to over-travel deflection plus a recovery wipe across the poppet, without the purge gap ([0083], [0087], [0146]); counsel weighs unity. · **Status:** addressed in fix pass (not re-reviewed)

#: 19 · **Deficiency or prior-art reference:** *(unclear)* Every product claim names a return spring. A poppet closed by chamber pressure alone, with the threshold set by ring stiffness, would escape, but no such embodiment is disclosed ([0043], [0144]). · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** high · **Source:** F-33-5 · **How to respond:** Ask the inventor whether a spring-less poppet was conceived and works; if so, disclose it before filing. · **Status:** open

#: 20 · **Deficiency or prior-art reference:** The only evidence of cleaning is on the wiped surfaces; film remained in the purge-gap region ([0182], [0190]). The effect of the purge path, the feature that distinguishes claim 1, is therefore unproven. · **Type:** Inventive step · **Claims affected:** 1, 58, 79, 81 · **Severity:** medium · **Source:** F-30-3 · **How to respond:** Add any observation or data that separates purge from wipe; explain the purge effect in [0192]; keep claims 25–26 as fallbacks. · **Status:** open

#: 21 · **Deficiency or prior-art reference:** With the purge effect unproven, an examiner can treat claim 1 as an alternative second flow path and combine two-threshold valves (EP 1137892 A1, US 8043186 B2) with a deflecting NiTi ring (US 9982818 B2). Motivation is thin. · **Type:** Inventive step · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-32-10 · **How to respond:** Add data isolating the air purge before filing or within the priority year; state the effect of the separate path; keep claims 25 and 47 as fallbacks. · **Status:** open

#: 22 · **Deficiency or prior-art reference:** Claim 1 covers every material, geometry and gap location, but only one NiTi species was tested, and purge and wipe were not separated. This invites a breadth attack (EPO

plausibility; UK breadth-of-claim insufficiency). · **Type:** Enablement · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-30-11 · **How to respond:** Obtain purge-versus-wipe data and a PEEK build if possible; make sure claims 11, 25 and 47 each stand as a fallback position. · **Status:** open

#: 23 · **Deficiency or prior-art reference:** The austenite finish temperature is claimed with no measurement method, and the value depends on the method used. · **Type:** Enablement · **Claims affected:** 48, 74, 95 · **Severity:** medium · **Source:** F-30-9 · **How to respond:** Add the test method and standard actually used for the prototype rings. · **Status:** open

#: 24 · **Deficiency or prior-art reference:** Claim 90 says the convolution may be "defined by the laser cutting", but cutting a tube does not move the wall radially, and Example 1 never says which step formed the convolution ([0169]). · **Type:** Enablement · **Claims affected:** 90, 93 · **Severity:** medium · **Source:** F-32-15 · **How to respond:** Record which step formed the convolution; delete "by the laser cutting" or explain it. · **Status:** open

#: 25 · **Deficiency or prior-art reference:** The worked-example dimensions do not visibly fit together: a 0.12 mm convolution on a 2.7 mm ring needs bore relief or a stem size that is never stated ([0111]). · **Type:** Enablement · **Claims affected:** 13, 16, 69, 78 · **Severity:** medium · **Source:** F-32-21 · **How to respond:** State the direction of the convolutions, any bore relief, and the poppet stem diameter. · **Status:** open

#: 26 · **Deficiency or prior-art reference:** The FIG. 1 instruction draws a clearance at rest, handing an opponent a picture of the purge gap open when claim 1 says it is closed. · **Type:** Support · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-28-9 · **How to respond:** Draw the free portion in contact at rest, or disclose what closes a running clearance to flow. · **Status:** open

#: 27 · **Deficiency or prior-art reference:** For the inner-surface gap of [0127], the main passage route is never described and "no significant flow" has no criterion. · **Type:** Support · **Claims affected:** 1, 31 · **Severity:** medium · **Source:** F-28-10 · **How to respond:** Describe the main-passage route and replace "no significant flow" with a statable criterion. · **Status:** open

#: 28 · **Deficiency or prior-art reference:** The "any one of claims" dependencies create combinations the description contradicts, e.g. claim 47 on claim 24, claim 34 on claim 29, and claim 99 on the PEEK ring claims. · **Type:** Support · **Claims affected:** 10, 13, 16, 17, 23, 24, 29, 34, 35, 41–43, 47–50, 76, 77, 99 · **Severity:** medium · **Source:** F-28-11 · **How to respond:** Prune the dependencies as listed in the finding; no supported scope is lost. · **Status:** open

#: 29 · **Deficiency or prior-art reference:** *(unclear)* The band-limited press fit of claim 21 is labelled in the package as a drafting assumption, not an inventor fact. · **Type:** Support · **Claims affected:** 21 · **Severity:** medium · **Source:** F-28-12 · **How to respond:** Inventor to confirm; if not confirmed, delete claim 21 and the matching passages. · **Status:** open

#: 30 · **Deficiency or prior-art reference:** With a single-rate spring and a stiffness-set threshold (claim 40), the ring is loaded before the threshold ([0144]), so it is not "at rest" in the normal-open state. For polymer rings the threshold is "less distinct". · **Type:** Support · **Claims affected:** 1, 40, 51–53 · **Severity:** medium · **Source:** F-28-13 · **How to respond:** Add inventor-confirmed wording that sub-threshold loading leaves the gap closed, or restrict claim 40 to the superelastic ring. · **Status:** open

#: 31 · **Deficiency or prior-art reference:** Fallbacks that pick one option from each of several lists rest only on the blanket statement in [0165], which the EPO and UK IPO rarely accept as a basis. · **Type:** Support · **Claims affected:** 1, 58, 67, 79 · **Severity:** medium · **Source:** F-29-5 · **How to respond:** Add a short "preferred combinations" section on the model of [0167]. · **Status:** open

#: 32 · **Deficiency or prior-art reference:** The three-tier ranges are separate lists; the intermediate or narrow tiers of different parameters are never stated together ([0111], [0112]). · **Type:** Support · **Claims affected:** 13, 16, 17, 35, 41–43, 67, 69, 70, 82 · **Severity:** medium · **Source:** F-29-6 · **How to respond:** Add sentences stating the intermediate tiers together and the narrow tiers together. · **Status:** open

#: 33 · **Deficiency or prior-art reference:** Same defect as row 9, limb on the range claims 16–17, 35, 41–43 and 68–70. · **Type:** Support · **Claims affected:** 16, 17, 35, 41–43, 67–70, 76, 77, 82 · **Severity:** medium · **Source:** F-30-20 · **How to respond:** As row 9. · **Status:** open

#: 34 · **Deficiency or prior-art reference:** Same defect as row 9, stated for PEEK in claims 76–77. · **Type:** Support · **Claims affected:** 16, 17, 35, 41–43, 67–70, 76, 77, 82 · **Severity:** medium · **Source:** F-32-17 · **How to respond:** As row 9. · **Status:** open

#: 35 · **Deficiency or prior-art reference:** In claim 31 supply air reaches the inner-surface gap only through the open main passage, so the purge path is not "separate from the main passage" as claim 1 requires. · **Type:** Support · **Claims affected:** 1, 31 · **Severity:** medium · **Source:** F-32-11 · **How to respond:** Reword claim 31 so that the gap is reached through the open main passage and is distinct from it. · **Status:** open

#: 36 · **Deficiency or prior-art reference:** The package contradicts itself: [0140] says the ring is "not touched" below the threshold, while [0144] requires contact below it. · **Type:** Support · **Claims affected:** 1, 40 · **Severity:** medium · **Source:** F-36-15 · **How to respond:** Amend [0140] to "not deflected so as to open the purge gap", with inventor confirmation; align claim 1 as in row 17. · **Status:** open

#: 37 · **Deficiency or prior-art reference:** Claimed features appear in no figure: integral flexure, return formation, inner-surface gap, openings in the wall, over-travel stop and depressor (37 CFR 1.83(a)). · **Type:** Support · **Claims affected:** 24, 28, 31, 32, 45, 46 · **Severity:** medium · **Source:** F-36-6 · **How to respond:** Add schematic partial sections in the style of FIGS. 11–14. · **Status:** open

#: 38 · **Deficiency or prior-art reference:** *(unclear)* A ring that flexes axially, not radially, to open the purge path escapes every "radial deflection" claim; no such embodiment is disclosed. · **Type:** Support · **Claims affected:** 1, 58, 67, 79 · **Severity:** medium · **Source:** F-36-9 · **How to respond:** If within the inventor's conception, disclose it and say the deformation may be radial or axial. · **Status:** open

#: 39 · **Deficiency or prior-art reference:** *(unclear)* No embodiment purges from first lift, with no normal stroke; every independent claim needs a normal-open state ([0073]). · **Type:** Support · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-36-10 · **How to respond:** If conceived, disclose it and add an independent claim without the normal-open state; counsel weighs EP 1137892 A1 and US 8043186 B2. · **Status:** open

#: 40 · **Deficiency or prior-art reference:** "Inflatable chamber" is defined as bicycle-only ([0046]), so e-scooter, motorcycle and other sealant tyres are outside claims 58 and 79. · **Type:** Support · **Claims affected:** 58, 79 · **Severity:** medium · **Source:** F-36-4 · **How to respond:** Redefine as any tyre or tube holding liquid sealant, with bicycles as the tested example. · **Status:** open

#: 41 · **Deficiency or prior-art reference:** Claim 64 claims the chamber as an article "comprising" a tube, while [0046] defines it as a volume. · **Type:** Support · **Claims affected:** 64 · **Severity:** medium · **Source:** F-37-3 · **How to respond:** Align [0046] with the claim, or the claim with [0046]. · **Status:** addressed in fix pass (not re-reviewed)

#: 42 · **Deficiency or prior-art reference:** The same volume-versus-article conflict makes claim 64 open to an indefiniteness attack; [0052] equates the chamber with the tube. · **Type:** Support · **Claims affected:** 64, 65 · **Severity:** medium · **Source:** F-38-1 · **How to respond:** Redefine [0046] as the physical article. · **Status:** open

#: 43 · **Deficiency or prior-art reference:** Claim 25 defines the product by a result on sealant film ("shears sealant film"), which is not part of the claim and is not measured. · **Type:** Clarity · **Claims affected:** 25, 80 · **Severity:** medium · **Source:** F-30-14 · **How to respond:** Recast kinematically: the ring moves across and in contact with the poppet surface during recovery. · **Status:** open

#: 44 · **Deficiency or prior-art reference:** Same result-language problem; against the separate wipers of US 4719938 A the distinction is the recovering ring, which the claim should recite. · **Type:** Clarity · **Claims affected:** 25, 80 · **Severity:** medium · **Source:** F-32-19 · **How to respond:** As row 43. · **Status:** open

#: 45 · **Deficiency or prior-art reference:** Clarity terms examiners will query: a "closed" gap with no criterion, "at-rest form", "low-adhesion", "tyre set-up", "surfaces on which sealant film accumulates". · **Type:** Clarity · **Claims affected:** 1, 25, 44, 49, 55, 64, 79, 95 · **Severity:** medium · **Source:** F-30-15 · **How to respond:** Reformulate each from existing description text, without changing scope. · **Status:** open

#: 46 · **Deficiency or prior-art reference:** No generic definition of a "closed" purge gap (the "no significant flow" test in [0127] covers one location only), and "to its at-rest form" conflicts with admitted residual set ([0121]). · **Type:** Clarity · **Claims affected:** 1, 58, 67, 79 · **Severity:** medium · **Source:** F-36-11 · **How to respond:** Add a

generic "closed" definition to [0039] and a "towards the at-rest form" wording to [0034]. · **Status:** open

#: 47 · **Deficiency or prior-art reference:** [0030] defines the key terms as "any structure capable of performing the recited function ... and their equivalents", inviting US means-plus-function treatment that confines claim 1 to the described structures. · **Type:** Clarity · **Claims affected:** 1, 2, 28, 36, 62 · **Severity:** medium · **Source:** F-30-1 · **How to respond:** Replace with structural definitions ([0035], [0042], [0043]) and state that means-plus-function treatment is not intended. · **Status:** open

#: 48 · **Deficiency or prior-art reference:** Same [0030] wording, applied to "deflecting formation" and "threshold-setting arrangement". · **Type:** Clarity · **Claims affected:** 1, 2, 36 · **Severity:** medium · **Source:** F-gate14-7 · **How to respond:** As row 47; counsel to consider structural wording in claim 36. · **Status:** open

#: 49 · **Deficiency or prior-art reference:** "Threshold-setting arrangement" pairs a nonce word with a function and is a likely US means-plus-function term; "deflecting formation" is close to the line. · **Type:** Clarity · **Claims affected:** 2, 36 · **Severity:** medium · **Source:** F-37-11 · **How to respond:** As row 47. · **Status:** open

#: 50 · **Deficiency or prior-art reference:** 102 claims, five independents, and multiple dependent claims resting on other multiple dependent claims, which the US forbids (37 CFR 1.75(c)); excess-claim fees in all three offices. · **Type:** Formal · **Claims affected:** 1, 10, 18, 58, 61, 64, 66, 67, 89, 99 · **Severity:** medium · **Source:** F-30-13 · **How to respond:** Restructure into single dependencies for the US; justify or fold the product independents; prune to the intended ladder. · **Status:** open

#: 51 · **Deficiency or prior-art reference:** The multiple-on-multiple dependency defect runs through 56 claims (US only). · **Type:** Formal · **Claims affected:** 10, 14, 18, 22–25, 27–29, 32–36, 40–47, 49–52, 54–57, 64, 66, 70–73, 75, 76, 78, 82–89, 93–96, 98–100, 102 · **Severity:** medium · **Source:** F-37-1 · **How to respond:** As row 50; no narrowing. · **Status:** addressed in fix pass (not re-reviewed)

#: 52 · **Deficiency or prior-art reference:** The claim structure is not adapted to US multiple-dependency rules or EPO Rule 43(2). · **Type:** Formal · **Claims affected:** 1, 10, 13, 58, 61, 64, 67, 89, 99 · **Severity:** medium · **Source:** F-gate14-3 · **How to respond:** Counsel prepares a single-dependency US set and the EPO interrelated-products justification. · **Status:** open

#: 53 · **Deficiency or prior-art reference:** Required description sections are missing (title, field, background, problem, summary, brief description of drawings). · **Type:** Formal · **Claims affected:** — · **Severity:** medium · **Source:** F-30-21 · **How to respond:** As row 11. · **Status:** open

#: 54 · **Deficiency or prior-art reference:** The figure plan carries internal references and statements of doubt ("not a confirmed mechanism", "the inventor states doubt") that must not reach the drawings. · **Type:** Formal · **Claims affected:** 1, 21, 30 · **Severity:** medium · **Source:** F-28-8 · **How to respond:** Strip them; write the brief description of the drawings from the description. · **Status:** open

#: 55 · **Deficiency or prior-art reference:** Same drafting-process content, plus a FIG. 1 instruction that cannot be drawn as written. · **Type:** Formal · **Claims affected:** — · **Severity:** medium · **Source:** F-32-5 · **How to respond:** Strip them; state one at-rest condition for FIG. 1. · **Status:** open

#: 56 · **Deficiency or prior-art reference:** The figure plan also has stale flags, and [0111] says dimensions are called out in FIG. 6A when the instructions call out none. · **Type:** Formal · **Claims affected:** — · **Severity:** medium · **Source:** F-gate14-8 · **How to respond:** Add the call-outs to FIG. 6A; re-run the numeral cross-check; unify the numeral style. · **Status:** open

#: 57 · **Deficiency or prior-art reference:** Disclosed alternatives outside the claims (purge actuator [0145], depressor-only [0088], inverted ring [0139], deflation-only [0089]): US dedication risk and literal escape. · **Type:** Design-around · **Claims affected:** 1, 46, 57, 58, 79 · **Severity:** medium · **Source:** F-30-2 · **How to respond:** As row 15. · **Status:** open

#: 58 · **Deficiency or prior-art reference:** A near-two-state design with the threshold close to the crack pressure ([0073]) sits at the edge of claim 1's normal-open state. · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-32-6 · **How to respond:** Consider an independent claim with the closed and purge states only. · **Status:** open

#: 59 · **Deficiency or prior-art reference:** [0073] contemplates a purge on substantially every opening; the variant with no real normal-open state is outside every independent claim. · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-33-1 · **How to respond:** Add an independent claim (core and method) with the closed and purge states only ([0035], [0073], [0087]). · **Status:** open

#: 60 · **Deficiency or prior-art reference:** The two-state, depressor-actuated and deflation-only modes ([0073], [0088], [0089]) have no independent claim. · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-36-16 · **How to respond:** As row 15. · **Status:** open

#: 61 · **Deficiency or prior-art reference:** Over-travel by depressor, by hand, during deflation only, or by a separate actuator falls outside claims 1, 58 and 79. · **Type:** Design-around · **Claims affected:** 1, 46, 58, 79 · **Severity:** medium · **Source:** F-gate14-10 · **How to respond:** As row 15; add a claim to the purge-actuator form ([0145]). · **Status:** open

#: 62 · **Deficiency or prior-art reference:** Claim 1 needs the ring "carried by the core body" and the poppet to cause the deflection; the purge actuator ([0145]) and the ring on the poppet ([0139]) escape. · **Type:** Design-around · **Claims affected:** 1, 58 · **Severity:** medium · **Source:** F-33-9 · **How to respond:** Add wording in which any member moved by the pressure differential causes the deflection, and the ring may be on the core body or the poppet. · **Status:** open

#: 63 · **Deficiency or prior-art reference:** "Carried by", read with [0026] (indirect carriage), arguably takes in a ring on the poppet, which widens the art against claim 1; [0139] cuts the other way. Scope is uncertain either way. · **Type:** Design-around · **Claims affected:** 1, 18, 24 · **Severity:** medium · **Source:** F-32-9 · **How to respond:** Recite the ring on one of core body and poppet and the deflecting formation on the other. · **Status:** open

#: 64 · **Deficiency or prior-art reference:** *(unclear)* A ring that flexes axially, or is pushed bodily off a land, escapes "radial deflection" ([0041]). · **Type:** Design-around · **Claims affected:** 1, 58, 67, 79 · **Severity:** medium · **Source:** F-33-13 · **How to respond:** If conceived, disclose it and claim "elastic deformation" without "radial"; counsel checks US 4408632 A and US 8043186 B2 first. · **Status:** open

#: 65 · **Deficiency or prior-art reference:** A permanent small clearance that carries no real flow is never "closed", so it escapes claims 1, 58 and 79. · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** medium · **Source:** F-33-14 · **How to respond:** As row 46. · **Status:** open

#: 66 · **Deficiency or prior-art reference:** Recovery "to" the at-rest form, and closing "below the over-travel threshold", are escaped by a ring with residual set ([0121]). · **Type:** Design-around · **Claims affected:** 1, 58, 67, 79 · **Severity:** medium · **Source:** F-33-17 · **How to respond:** Recite recovery "towards" the at-rest form and closing "as the pressure differential falls". · **Status:** open

#: 67 · **Deficiency or prior-art reference:** Superelastic hysteresis can keep the gap open below the threshold, so the closing condition of claims 1 and 58 is not met literally ([0096]). · **Type:** Design-around · **Claims affected:** 1, 40, 58 · **Severity:** medium · **Source:** F-32-14 · **How to respond:** Use claim 79's wording: the gap closes as the pressure differential falls. · **Status:** open

#: 68 · **Deficiency or prior-art reference:** The wipe-only embodiment ([0087]) has no claim family. · **Type:** Design-around · **Claims affected:** 1, 25, 79, 80 · **Severity:** medium · **Source:** F-36-3 · **How to respond:** As row 18, with a matching method claim. · **Status:** open

#: 69 · **Deficiency or prior-art reference:** Claim 67 is limited to Presta-sized rings, so Schrader-scale rings ([0159]) and rings just outside the window escape. · **Type:** Design-around · **Claims affected:** 67 · **Severity:** medium · **Source:** F-33-6 · **How to respond:** Consider a dimension-free ring claim, keeping claims 16, 17 and 78 as fallbacks; weigh against row 13. · **Status:** open

#: 70 · **Deficiency or prior-art reference:** Claim 67 is also escaped by a wall convoluted along its whole length, because [0049] measures diameter at a convolution-free portion. · **Type:** Design-around · **Claims affected:** 67 · **Severity:** medium · **Source:** F-33-15 · **How to respond:** Add a fallback measurement convention to [0049], e.g. the diameter at the sealing edge. · **Status:** open

#: 71 · **Deficiency or prior-art reference:** A matched ring, poppet and spring kit sold without a core body infringes no product claim directly. · **Type:** Design-around · **Claims affected:** 1, 67 · **Severity:** medium · **Source:** F-33-18 · **How to respond:** Add a kit claim ([0163], FIG. 5). · **Status:** open

#: 72 · **Deficiency or prior-art reference:** The kit has no claim and the component claim is dimension-bound. · **Type:** Design-around · **Claims affected:** 1, 67 · **Severity:** medium · **Source:** F-gate14-9 · **How to respond:** As row 71. · **Status:** open

#: 73 · **Deficiency or prior-art reference:** No claims to non-bicycle sealant chambers, the kit, or the valve with a depressor head or adaptor ([0088], [0161], [0163]). · **Type:** Design-around · **Claims affected:** 64, 65, 66 · **Severity:** medium · **Source:** F-36-8 · **How to respond:** Add a generic chamber claim, a kit claim and an inflation-system claim. · **Status:** open

#: 74 · **Deficiency or prior-art reference:** Claim 90 is performed in the supplier's factory and escaped by air cooling or by photochemical etching ([0173]). · **Type:** Design-around · **Claims affected:** 90 · **Severity:** medium · **Source:** F-33-11 · **How to respond:** Move water quenching and laser cutting into dependent claims; counsel weighs EPO intermediate generalisation. · **Status:** open

#: 75 · **Deficiency or prior-art reference:** Method claim 79 is performed by the cyclist, and infringement depends on proving sealant film on the accused surfaces. · **Type:** Design-around · **Claims affected:** 79, 81 · **Severity:** medium · **Source:** F-33-12 · **How to respond:** Recite "across a surface of the deflectable ring element" ([0078]); move the film wording to a dependent claim. · **Status:** open

#: 76 · **Deficiency or prior-art reference:** [0199] calls the stainless-steel core body a matter of general materials knowledge, which is a ready obviousness admission against claim 54. · **Type:** Inventive step · **Claims affected:** 54 · **Severity:** low · **Source:** F-38-3 · **How to respond:** Reword [0199] to describe the corrosion consideration without the admission. · **Status:** open

#: 77 · **Deficiency or prior-art reference:** No design relationship links over-travel, convolution amplitude and strain; the key values were design values, not measurements ([0110], [0113]). · **Type:** Enablement · **Claims affected:** 13, 35, 43 · **Severity:** low · **Source:** F-30-17 · **How to respond:** Add the relationship used, or state that the over-travel stop limits the ring load. · **Status:** open

#: 78 · **Deficiency or prior-art reference:** The Schrader variant has no operating sequence for a pin that every pump head presses ([0159]). · **Type:** Enablement · **Claims affected:** 46, 57 · **Severity:** low · **Source:** F-36-13 · **How to respond:** Add the Schrader operating sequence and the depressor travel. · **Status:** open

#: 79 · **Deficiency or prior-art reference:** *(unclear)* Behaviour on top-up inflation from riding pressure is not described; every test ran from fully deflated. · **Type:** Enablement · **Claims affected:** 1, 79, 87 · **Severity:** low · **Source:** F-36-5 · **How to respond:** Add after [0091] whether the purge state is reached on top-up, or say that it is unknown. · **Status:** open

#: 80 · **Deficiency or prior-art reference:** *(unclear)* Any test data beyond Example 1 (more cycles, cold, PEEK, wiper ring) is not in the text ([0182], [0188]). · **Type:** Enablement · **Claims affected:** 1 · **Severity:** low · **Source:** F-36-17 · **How to respond:** Ask the inventor; add any results as actual examples. · **Status:** open

#: 81 · **Deficiency or prior-art reference:** "Defined wiping edge" is disclosed only as the laser-cut sharp edge ([0149]), yet claimed for any ring. · **Type:** Support · **Claims affected:** 27 · **Severity:** low · **Source:** F-28-14 · **How to respond:** Define it in the definitions and extend it to other rings only if the inventor agrees. · **Status:** open

#: 82 · **Deficiency or prior-art reference:** Some claimed combinations are supported only implicitly, e.g. the 0.03 mm interference with a polymer ring. · **Type:** Support · **Claims affected:** 9, 18, 20, 26, 59, 63, 102 · **Severity:** low · **Source:** F-28-19 · **How to respond:** Add one confirming sentence per combination, or prune. · **Status:** open

#: 83 · **Deficiency or prior-art reference:** Blanket statements ([0027], [0165]) will not secure an amendment basis at the EPO or in the UK. · **Type:** Support · **Claims affected:** 1 · **Severity:** low · **Source:** F-30-10 · **How to respond:** Add explicit fallback paragraphs from existing content (as row 31). · **Status:** open

#: 84 · **Deficiency or prior-art reference:** No ranges for the normal stroke, the shoulder's rest clearance or the spring-rate ratio ([0067], [0043]). · **Type:** Support · **Claims affected:** 2, 37, 41, 43 · **Severity:** low · **Source:** F-36-12 · **How to respond:** Ask the inventor for ranges they conceived; do not generate them. · **Status:** open

#: 85 · **Deficiency or prior-art reference:** No ranges for the normal stroke, the interference or the shape-setting window ([0170], [0195], [0198]). · **Type:** Support · **Claims affected:** 20, 94, 101 · **Severity:** low · **Source:** F-gate14-11 · **How to respond:** As row 84. · **Status:** open

#: 86 · **Deficiency or prior-art reference:** *(unclear)* No narrower Af value to retreat to if cold performance becomes the battleground ([0095]). · **Type:** Support · **Claims affected:** 48, 74 · **Severity:** low · **Source:** F-36-19 · **How to respond:** Add a nested ladder only if the inventor has chosen values. · **Status:** open

#: 87 · **Deficiency or prior-art reference:** *(unclear)* No basis to amend to recovery speed, force or the audible snap ([0099], [0181]). · **Type:** Support · **Claims affected:** 1, 25, 47 · **Severity:** low · **Source:** F-29-9 · **How to respond:** Add a measured recovery time or force if one exists; otherwise rely on the material claims. · **Status:** open

#: 88 · **Deficiency or prior-art reference:** The diameter convention of [0049] fails for a ring convoluted end to end. · **Type:** Support · **Claims affected:** 16, 67 · **Severity:** low · **Source:** F-36-21 · **How to respond:** As row 70. · **Status:** open

#: 89 · **Deficiency or prior-art reference:** [0100] ("no heating") reads against the thermally actuated purge of [0164]. · **Type:** Support · **Claims affected:** — · **Severity:** low · **Source:** F-38-2 · **How to respond:** Confine [0100] to the over-travel embodiments. · **Status:** open

#: 90 · **Deficiency or prior-art reference:** The inverted ring and the purge actuator have no fallback claim ([0139], [0145]). · **Type:** Support · **Claims affected:** 1, 58 · **Severity:** low · **Source:** F-36-18 · **How to respond:** As row 62. · **Status:** open

#: 91 · **Deficiency or prior-art reference:** *(unclear)* A spring-less poppet closed by chamber pressure is not disclosed. · **Type:** Support · **Claims affected:** 1, 58, 79 · **Severity:** low · **Source:** F-36-20 · **How to respond:** As row 19. · **Status:** open

#: 92 · **Deficiency or prior-art reference:** "A purge gap associated with the deflectable ring element" states no physical relationship. · **Type:** Clarity · **Claims affected:** 1, 58, 79 · **Severity:** low · **Source:** F-30-19 · **How to respond:** Replace with "bounded at least in part by, or formed through, the deflectable ring element" ([0124]–[0129]). · **Status:** open

#: 93 · **Deficiency or prior-art reference:** "About", "approximately" and "nominally" in claims, while [0028] says they define no tolerance. · **Type:** Clarity · **Claims affected:** 20, 44, 49, 75, 98, 101 · **Severity:** low · **Source:** F-30-23 · **How to respond:** Recast claim 44 on the sub-ranges of [0112]; get tolerances from the inventor or supplier for the rest. · **Status:** open

#: 94 · **Deficiency or prior-art reference:** Claim 1's closing shape-recovery clause is not framed parallel to the three states. · **Type:** Clarity · **Claims affected:** 1 · **Severity:** low · **Source:** F-37-5 · **How to respond:** Add a lead-in phrase. · **Status:** open

#: 95 · **Deficiency or prior-art reference:** "The chamber side" is applied to the fixed seat and the purge gap, but [0031] defines it only for the poppet. · **Type:** Clarity · **Claims affected:** 9, 33 · **Severity:** low · **Source:** F-37-10 · **How to respond:** Broaden [0031] to a general orientation. · **Status:** open

#: 96 · **Deficiency or prior-art reference:** [0183] cross-refers to a background that is not in the package. · **Type:** Formal · **Claims affected:** 1 · **Severity:** low · **Source:** F-28-15 · **How to respond:** Re-point [0183] to [0045] and [0050]. · **Status:** open

#: 97 · **Deficiency or prior-art reference:** Wording drift: "not exceeding" against "below" ([0064]); "to" against "towards" ([0034]); numeral 30 used for ring 730 in claim 58. · **Type:** Formal · **Claims affected:** 1, 58 · **Severity:** low · **Source:** F-28-20 · **How to respond:** Align the wording and numerals. · **Status:** open

#: 98 · **Deficiency or prior-art reference:** Numeral 30 is attached to the genus in claims 1 and 58, and numeral 34 is reused for the fixed seat's sealing edge. · **Type:** Formal · **Claims affected:** 1, 58 · **Severity:** low · **Source:** F-30-25 · **How to respond:** Drop or widen the numeral; give the fixed seat's edge its own numeral. · **Status:** open

#: 99 · **Deficiency or prior-art reference:** Claim 58 uses (30) for ring 730; claims 100–102 recite assembly under a ring-manufacturing preamble. · **Type:** Formal · **Claims affected:** 58, 100–102 · **Severity:** low · **Source:** F-30-26 · **How to respond:** Claim 58 "(30, 730)"; retitle claims 100–102 as making a valve core. · **Status:** open

#: 100 · **Deficiency or prior-art reference:** Numeral format drift ([0051]–[0062] unbracketed), registry mismatches, and the claim 100 preamble. · **Type:** Formal · **Claims affected:** 58, 100–102 · **Severity:** low · **Source:** F-32-22 · **How to respond:** Bracket numerals uniformly; correct the registry rows. · **Status:** open

#: 101 · **Deficiency or prior-art reference:** The genus "deflectable ring element (30)" carries the seat-ring numeral, although claim 9 makes the wiper ring (132) the element. · **Type:** Formal · **Claims affected:** 1, 8, 9 · **Severity:** low · **Source:** F-37-7 · **How to respond:** Delete the numeral from the genus. · **Status:** addressed in fix pass (not re-reviewed)

#: 102 · **Deficiency or prior-art reference:** Claims 79–89 carry no reference numerals, unlike the other independents. · **Type:** Formal · **Claims affected:** 79–89 · **Severity:** low · **Source:** F-37-8 · **How to respond:** Add numerals once the drawings exist. · **Status:** open

#: 103 · **Deficiency or prior-art reference:** Same reference-sign inconsistency across the claim set. · **Type:** Formal · **Claims affected:** 79–89 · **Severity:** low · **Source:** F-40a-3 · **How to respond:** As row 102. · **Status:** addressed in fix pass (not re-reviewed)

#: 104 · **Deficiency or prior-art reference:** *(unclear)* The independent claims are not in EPO two-part form; whether that is "appropriate" depends on the closest art found by the search report. · **Type:** Formal · **Claims affected:** 1, 58, 67, 79, 90 · **Severity:** low · **Source:** F-40a-4 · **How to respond:** Decide at EP and UK entry, after the search report. · **Status:** open

#: 105 · **Deficiency or prior-art reference:** No statement that reference signs in the claims are non-limiting. · **Type:** Formal · **Claims affected:** — · **Severity:** low · **Source:** F-40a-5 · **How to respond:** Add the standard sentence at [0001]. · **Status:** open

#: 106 · **Deficiency or prior-art reference:** "Return spring" is the species; [0030] names the genus "biasing member". · **Type:** Design-around · **Claims affected:** 1, 58, 79 · **Severity:** low · **Source:** F-33-19 · **How to respond:** Use "a biasing member (50)" in the independents ([0030], [0043]). · **Status:** open

#: 107 · **Deficiency or prior-art reference:** Method claim 79 is performed by the end user and is confined to bicycle chambers by [0046]. · **Type:** Design-around · **Claims affected:** 79 · **Severity:** low · **Source:** F-33-21 · **How to respond:** Keep product claims free of in-use requirements; see row 40. · **Status:** open

#: 108 · **Deficiency or prior-art reference:** A depressor adaptor gives over-travel without "raising the pressure differential"; claim 79 is then not performed. · **Type:** Design-around · **Claims affected:** 79 · **Severity:** low · **Source:** F-35-6 · **How to respond:** Add a method claim with over-travel by mechanical depression ([0088], [0089]). · **Status:** open

#: 109 · **Deficiency or prior-art reference:** US 9982818 B2 (UTC Aerospace Systems, 2018): a NiTi sealing ring with a wave-like section giving radial displacement. It is a static, thermally actuated seal, with no poppet, no sealing edge and no superelastic recovery. Closest art for the ring shape and material. · **Type:** Prior art · **Claims affected:** 10–13, 47, 67–78, 90 · **Severity:** high · **Source:** US 9982818 B2 · **How to respond:** Rely on the moving-poppet seat, over-travel and purge; narrow claim 67 as in row 13. · **Status:** prior art

#: 110 · **Deficiency or prior-art reference:** US 6393765 B1 (US Navy, 2002): superelastic NiTi seal elements that deform on closure and recover when opened (hatches, doors). Shows superelastic NiTi was known as a repeatedly deforming seal. · **Type:** Prior art · **Claims affected:** 47–50, 67, 73–75 · **Severity:** high · **Source:** US 6393765 B1 · **How to respond:** Combination partner for row 109; answer with the valve-seat role and the over-travel purge. · **Status:** prior art

#: 111 · **Deficiency or prior-art reference:** US 11904644 B2 (Santa Cruz Bicycles, 2024): a tubeless valve made because Presta and Schrader cores clog with sealant; a resilient seat with annular flow around it. No deforming seat, over-travel or wipe. Likely primary reference for the problem. · **Type:** Prior art · **Claims affected:** 1, 58, 64–66, 79, 85 · **Severity:** medium · **Source:** US 11904644 B2 · **How to respond:** Distinguish on the deflecting ring, the separate purge path and recovery. · **Status:** prior art

#: 112 · **Deficiency or prior-art reference:** US 4719938 A (Helix Technology, 1988): wiper rings, pushed outward by a spring ring, wipe debris from the sealing surface as the valve closes. A combination partner against the wipe. · **Type:** Prior art · **Claims affected:** 25–27, 80 · **Severity:** medium · **Source:** US 4719938 A · **How to respond:** Recite the wipe by recovery of the seat ring itself (row 43). · **Status:** prior art

#: 113 · **Deficiency or prior-art reference:** US 11345196 B2 (Kunshan Fuluo, 2022): tyre valve where a seal ring on a movable rod squeezes repair liquid off the wall when the rod is reinserted. Wiping against sealant, but by a maintenance movement. · **Type:** Prior art · **Claims affected:** 25–27, 80 · **Severity:** medium · **Source:** US 11345196 B2 · **How to respond:** Distinguish on automatic wiping by seat recovery. · **Status:** prior art

#: 114 · **Deficiency or prior-art reference:** US 4462449 A (Zabel, 1984): tyre valve core in which soil is "swept through" on inflation and the core is "self cleaning". Early inflation-flow self-cleaning; no seat ring or wipe. · **Type:** Prior art · **Claims affected:** 1, 79, 81 · **Severity:** medium · **Source:** US 4462449 A · **How to respond:** Distinguish on the pressure-threshold purge path opened by ring deflection. · **Status:** prior art

#: 115 · **Deficiency or prior-art reference:** US 8043186 B2 (Schaeffler, 2011): one check valve with two spring-set opening pressures, the second opening a second annular flow path. Paths are opened by separate members, not by a deflecting seat. A combination partner against the normal-stroke-then-over-travel sequence. · **Type:** Prior art · **Claims affected:** 1, 36–39, 58, 79 · **Severity:** medium · **Source:** US 8043186 B2 · **How to respond:** Distinguish on one poppet deflecting a resilient ring that recloses by shape recovery. · **Status:** prior art

#: 116 · **Deficiency or prior-art reference:** EP 1137892 A1 (Sumnett, 2001): tyre valve core with a calibrated spring; above a set pressure the disk exposes bleed apertures, a second path in a tyre core opened above a threshold. Claim 1 not yet charted. · **Type:** Prior art · **Claims affected:** 1, 36, 58 · **Severity:** medium · **Source:** EP 1137892 A1 · **How to respond:** Chart claim 1 from the saved text; distinguish on ring deflection and supply-to-chamber purge. · **Status:** prior art

#: 117 · **Deficiency or prior-art reference:** US 4408632 A (Brunswick, 1983): poppet with concentric primary and secondary seats opening paths in sequence as it travels. Background for a later-stage second path. · **Type:** Prior art · **Claims affected:** 1 · **Severity:** low · **Source:** US 4408632 A · **How to respond:** Background only on the saved text. · **Status:** prior art

#: 118 · **Deficiency or prior-art reference:** US 4627461 A (K.J. Baillie, 1986): the closure member is rotated against the seat on closing to grind deposits away. An alternative seat-cleaning mechanism. · **Type:** Prior art · **Claims affected:** 25–27 · **Severity:** low · **Source:** US 4627461 A · **How to respond:** Distinguish on recovery wiping without rotation. · **Status:** prior art

#: 119 · **Deficiency or prior-art reference:** US 6006954 A (Bespak, 1999): a wiper on a metering-valve stem scrapes deposits from the chamber wall on both strokes. · **Type:** Prior art · **Claims affected:** 25–27, 80 · **Severity:** low · **Source:** US 6006954 A · **How to respond:** As row 112. · **Status:** prior art

#: 120 · **Deficiency or prior-art reference:** CN 215861914 U (Jinan Weidisi, 2022): exhaust air flow blows debris off a filter beside the sealing gasket in a tyre valve; read in machine translation. · **Type:** Prior art · **Claims affected:** 79, 81, 88 · **Severity:** low · **Source:** CN 215861914 U · **How to respond:** Distinguish on the purge path through a deflected ring. · **Status:** prior art

#: 121 · **Deficiency or prior-art reference:** US 12215796 B2 (Santa Cruz Bicycles, 2025): an external seat and a resilient seating surface on a through-stem. No deforming seat, over-travel or purge. · **Type:** Prior art · **Claims affected:** 1, 58 · **Severity:** low · **Source:** US 12215796 B2 · **How to respond:** Background; shows the field solves clogging by enlarging the seat. · **Status:** prior art

#: 122 · **Deficiency or prior-art reference:** US 11179979 B2 (Sport Components, 2021): a duckbill valve at the stem base that keeps sealant out. Exclusion, not removal. Claim 1 not yet charted. · **Type:** Prior art · **Claims**

affected: 58, 64, 65 · **Severity:** low · **Source:** US 11179979 B2 · **How to respond:** Background; chart claim 1 from the saved text. · **Status:** prior art

#: 123 · **Deficiency or prior-art reference:** US 11065925 B2 (Liquitube, 2021): clog-resistant stem with an extension tube and side air holes. Same problem, flow-path solution. · **Type:** Prior art · **Claims affected:** 58, 64 · **Severity:** low · **Source:** US 11065925 B2 · **How to respond:** Background. · **Status:** prior art

#: 124 · **Deficiency or prior-art reference:** EP 4735276 A1 (applicant not identified, published 6 May 2026): replaces the Presta core with a ball valve to cure clogging. Published before any filing here, so it is full prior art. · **Type:** Prior art · **Claims affected:** 1, 58 · **Severity:** low · **Source:** EP 4735276 A1 · **How to respond:** Background; take the applicant from the saved text. · **Status:** prior art

#: 125 · **Deficiency or prior-art reference:** US 9297373 B2 (Nuovo Pignone, 2016): poppet with two parallel non-linear springs. Background for a non-linear return spring; not sequential stages. · **Type:** Prior art · **Claims affected:** 37 · **Severity:** low · **Source:** US 9297373 B2 · **How to respond:** Background. · **Status:** prior art

#: 126 · **Deficiency or prior-art reference:** Intake test document ref-01: sealant-resistant core with removable insert, low-adhesion poppet, rigid seat. The file states it and its number are fictional. · **Type:** Prior art · **Claims affected:** 1, 55 · **Severity:** low · **Source:** intake/prior-art/ref-01-sealant-resistant-core.md · **How to respond:** Chart from the intake file only; do not cite a number. · **Status:** prior art

#: 127 · **Deficiency or prior-art reference:** Intake test document ref-02: elastomer lip seat that deflects outward under inflation pressure (main path only). The file states it and its number are fictional. · **Type:** Prior art · **Claims affected:** 1, 8, 52 · **Severity:** low · **Source:** intake/prior-art/ref-02-elastomer-seat-valve.md · **How to respond:** Chart from the intake file only; distinguish on the separate purge path opened only in over-travel. · **Status:** prior art

#: CN-1 · **Deficiency or prior-art reference:** claims 4, 5 / purge shoulder (44) on the chamber side of the sealing face, bearing axially on the deflectable ring element in over-travel / the axial order of internal shoulder (24), sealing edge (34), convolutions (32a, 32b) and purge shoulder (44), and the ring surface the purge shoulder bears on, from the inventor's FIG-1 sketch; [0059], [0074], [0101], [0138] reconciled (F-28-3, F-32-20, F-36-1, F-gate14-2) · **Type:** support (claims need disclosure) · **Claims affected:** 4 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-2 · **Deficiency or prior-art reference:** claim 22 / interference fit confined to an axial band adjacent the internal shoulder, remainder free of the core body bore / the route by which the purge flow in the outer-surface purge gap passes the interference band to reach the chamber side in the embodiment of FIGS. 1 to 4 (F-28-3, F-36-1) · **Type:** support (claims need disclosure) · **Claims affected:** 22 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-3 · **Deficiency or prior-art reference:** claim 33 / purge gap bounded by the inner surface of the seat ring and the poppet on the chamber side of the sealing edge, flow along the poppet stem / placement of this gap consistent with [0110] (ring bore admits only the poppet stem) (F-28-3) · **Type:** support (claims need disclosure) · **Claims affected:** 33 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-4 · **Deficiency or prior-art reference:** claims 30, 31, 32 / sense of radial deflection (outward or inward at the bounding surface) and the outer-surface purge gap / the prototype's actual kinematics: moving surface, sense, purge gap location; [0041], [0061], [0075], [0076] and the FIG. 1, 3, 6 instructions conformed, and the closed-state clearance removed from FIG. 1 (F-28-2, F-30-6, F-gate14-4) · **Type:** support (claims need disclosure) · **Claims affected:** 30 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-5 · **Deficiency or prior-art reference:** claims 10, 84 / wiper ring deflected outward opening an outer-surface purge gap; seat ring deflecting radially outward / same kinematics confirmation for the wiper ring embodiment; [0115], [0147], [0185] and the FIG. 10B instruction conformed (F-28-2, F-gate14-4) · **Type:** support (claims need disclosure) · **Claims affected:** 10 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-6 · **Deficiency or prior-art reference:** claims 1, 61, 69, 92 / normal-open state without the at-rest form of the deflectable ring element / sentence in [0140] and [0144] that the deflecting formation may contact and elastically load the ring below the over-travel threshold without opening the purge gap; 'is not touched' amended to 'is not deflected so as to open the purge gap'; inventor confirmation needed (F-33-2) · **Type:** support (claims need disclosure) · **Claims affected:** 1 · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

#: CN-7 · **Deficiency or prior-art reference:** all claims / reference numerals (10, 12, 20, 22, 24, 30, 32, 34, 40, 41, 42, 44, 50, 52, 54, 60, 62, 64, 130, 132, 224, 232, 244, 254, 262, 324, 332, 344, 354, 362, 432, 444, 620, 630, 730) / drawing sheets, brief description of the drawings and abstract; without sheets the numerals point at nothing and figure-based support is at risk (F-gate14-1) · **Type:** support (claims need disclosure) · **Claims affected:** see item · **Severity:** medium · **Source:** claims fix pass · **How to respond:** add supporting disclosure before filing, or amend the claim · **Status:** open

How the score was reached

- **Novelty: 2 / 2.** No saved reference discloses every feature of claim 1. The index finds nothing with a superelastic seat against which a moving member seals, nothing with the over-travel-plus-purge combination in any valve, and nothing with seat-recovery wiping. US 9982818 B2 does not anticipate claim 67 (static, thermally actuated seal; row 109).
- **Inventive step: 1 / 2.** For claims 67 and 90 a credible combination exists (US 9982818 B2 with US 6393765 B1; rows 13–14). Claim 1 is arguable: its purge effect is not isolated by the data ([0182], [0190]; rows 20–21).
- **Support and enablement: 1 / 2.** Open high findings and no critical one. The built embodiment's layout and purge mechanism are contradictory or missing (rows 1–8), and the ranges are tied to NiTi (row 9). Claim 1 survives on [0125]–[0128].
- **Clarity and form: 1 / 2.** Open high formal findings (no drawings or abstract, unity; rows 10–12) and no critical one. The US dependency structure must be rebuilt (rows 50–52).
- **Eligibility and fallbacks: 1 / 2.** A mechanical valve, eligible in the US, EP and GB. The claim ladder is deep (claims 2–57, 79–102), but at the EPO and UK IPO multi-list fallbacks rest on the blanket statement in [0165] (rows 31, 83), and the range fallbacks are tied to one material (row 9).

Per jurisdiction:

- **US: 7 / 10.** Fallbacks score 2: the dependency defects are cured by preliminary amendment. Dedication of unclaimed alternatives (rows 15–18) costs scope, not grant.
- **EP: 6 / 10.** Added-matter practice makes the missing layout unrepairable after filing and limits the fallback combinations; unity and Rule 43(2) apply to claims 1, 58, 67 and 90.
- **GB: 6 / 10.** Added-matter practice (s.76) and breadth-of-claim sufficiency apply as at the EPO.

The score rests only on the 17 references we found and saved (coverage is thin: web-only, English-language, no CPC walk of F16K 15/20 or B60C 29/00). A professional search could change it. It is the application as reviewed in round 1; the fix pass running in parallel is not scored.

AI-assisted estimate for review by a qualified patent attorney or agent. Not legal advice and not a patentability opinion. Based only on the prior art identified in this matter.